

Kinetically Controlled Self-Assembly of Phthalocyanine–Peptide Conjugate Nanofibrils Enabling Superlarge Redshifted Absorption

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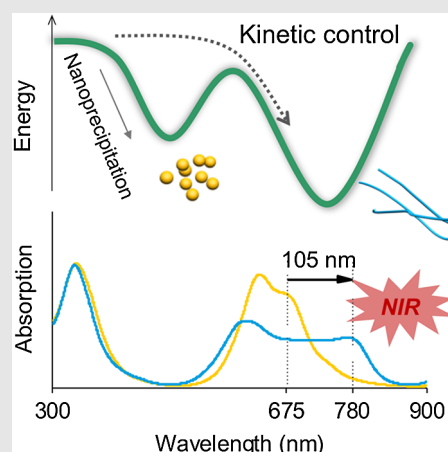
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Cite this: *CCS Chem.* **2019**, *1*, 173–180

Supramolecular assembly could in principle lead to redshifted absorption through *J*-aggregation of chromophores, which would be a highly promising method for achieving near-infrared materials with improved functionality and flexibility. To effectively enhance the material functionalities, one of the great challenges remaining is to achieve an aggregation state with a redshift larger than 100 nm. Conventional assemblies that are mostly thermodynamically controlled have a limited redshifted absorption of less than 30 nm. In this work, using a model phthalocyanine–peptide conjugate compound, we achieved the first fabrication of phthalocyanine-based near-infrared materials with a superlarge absorption redshift of 105 nm by a kinetically controlled self-assembly strategy. In this kinetically controlled self-assembly process, sufficient rearrangement of intermolecular aggregates to an ordered structure is revealed to be crucial to facilitate the formation of nanofibrils instead of nanoparticles, which are formed via a general rapid self-assembly pathway under thermodynamic control. The superlarge redshift in the absorbance of assembled nanofibrils originates from the orderly stacked phthalocyanine chromophores,

which enable a charge transfer state through more effective intermolecular orbital overlapping. The kinetically controlled *J*-aggregation state of the phthalocyanine–peptide conjugate with superlarge redshifted absorption not only opens an unprecedented route toward novel near-infrared phthalocyanine materials but also holds great promise for revealing general design principles for various organic dye materials.



Keywords: phthalocyanine, peptide, self-assembly, redshift absorption, nanofibers

Organic dye materials with near-infrared absorption are highly attractive in photoelectric devices, optical diagnostics, and therapeutics due to their inherent advantages of low cost, narrow and tunable energy levels, and high compatibility.^{1–6} Supramolecular assembly of simple organic dye molecules for the construction of near-infrared materials may lead to redshifted absorption through *J*-aggregation of chromophores, which has the potential to achieve absorption in the near-infrared region in a way that is much more convenient and flexible than conventional molecular synthesis.^{7–9} Ideally, obtaining materials with near-infrared absorption requires a redshift larger than 100 nm. However, compared with their nonaggregated form, most of the reported self-assembled organic materials have shown a redshift of less than 30 nm in absorption, resulting in limited variance in properties and functionalities. The failure of general supramolecular assembly routes in realizing such a superlarge redshift is likely because these supramolecular materials are fabricated without precise modulation of weak intermolecular interactions, while some potential metastable aggregation states with superlarge redshifted absorption are difficult to achieve.

Inspired by natural nonequilibrium systems with unique structures and functions, kinetic assembly may provide metastable nanostructures that are otherwise

thermodynamically inaccessible due to the assembly pathway complexity of these supramolecular systems.^{10–16} The key for kinetic assembly is to tightly control the noncovalent intermolecular interactions in the dimensions of both time and space via several parameters, including temperature, external electric or magnetic field, solvent pH, and order of addition.¹⁷ With suitable kinetic methodologies, different aggregate morphologies and molecular arrangements can be rationally selected from the same starting building blocks. This ability is crucial for developing supramolecular materials with optimized functional properties.^{18–22} To date, many kinetic assembly modulated materials with intriguing functionalities have been fabricated, but kinetic assembly for achieving a superlarge redshift in absorption has still rarely been reported. In addition, how a superlarge redshift is generated from a specific aggregation state has not yet been elaborated. As a consequence, the discovery of an efficient supramolecular assembly route is vital for creating near-infrared materials with superlarge redshifted absorption and clarifying the inherent mechanism.

In this study, we employed a model molecular compound, phthalocyanine-diphenylalanine (PF) conjugate, as the building block to achieve the superlarge redshifted absorption (Figure 1a). Phthalocyanine is an excellent

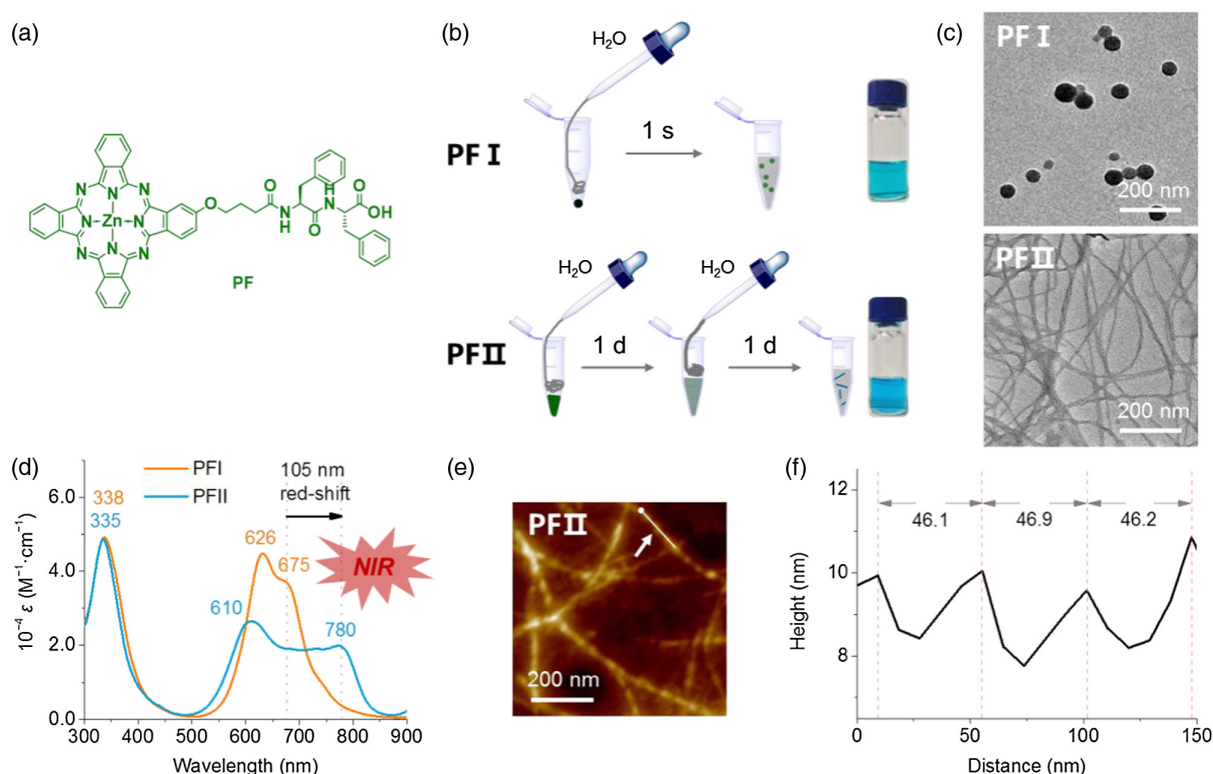


Figure 1 | Schematic illustration of (a) the molecular structure of peptide-conjugated zinc phthalocyanine, (b) two assembly processes, (c) TEM images of PFI and PFII, (d) electronic absorption spectra of PFI and PFII with a superlarge redshift for the latter, and (e, f) AFM image and section of PFII.

self-assembly building block due to its highly symmetric molecular structure, high flexibility of modification, and strong intermolecular π - π interactions.^{23,24} Because of these unique advantages, phthalocyanine shows potential in an extremely wide range of applications, such as organic semiconductors,²⁵ single-molecule magnets,²⁶ green catalysts,²⁷ and phototherapeutic agents.^{28,29} On the other hand, the conjugated peptide diphenylalanine is a natural amphiphilic motif that has remarkable biological functions and is able to tune the assembly of various functional molecules to obtain nanomaterials with controllable morphology and considerable stability in an aqueous medium.^{30–34} The pure PF compound was synthesized through four steps (see [Supporting Information Scheme S1](#)), and its molecular structure and purity were characterized by ¹H NMR and HR-MALDI-TOF MS (see [Supporting Information Figures S1–S3](#)).

As a general self-assembly method, a rapid mixing of water and a concentrated dimethyl sulfoxide (DMSO) solution of PF afforded stable nanoparticles (PFI) with particle sizes of approximately 50 nm and absorption peaks at 630 and 675 nm (Figure 1b–d). In comparison with the intense absorption at 675 nm of the monomeric PF solution in DMSO, the absorption of PFI exhibited no redshift (see [Supporting Information Figure S4](#)). Alternatively, several kinetic assembly methodologies were attempted, including one route that increased the initial volume ratio of DMSO/water solvent mixture from 1:80 (the preparation condition of PFI) to 1:5, followed by elongating the PF assembly time by slowly adding water many times over 3 days. This specific kinetic assembly procedure led to a colloid system with a blue color that differed from the cyanine of PFI (Figure 1b). Interestingly, the absorption of the kinetically prepared nanostructures expanded to the near-infrared region with a redshift of larger than 100 nm from 675 to 780 nm (Figure 1d). The change in the absorption spectrum indicated a distinct assembly structure, which was denoted PFII. Transmission electron microscopy (TEM) images showed that PFII consisted of soft helical fibrils with diameters of 8–12 nm for a single fibril and a screw pitch of approximately 45 nm, which was completely distinct from the PFI nanoparticles (Figure 1c; see [Supporting Information Figure S5a](#)). Atomic force microscopy (AFM) images further confirmed the helical nanofibril structure of PFII with diameters of approximately 9 nm and an average screw pitch of 46.5 nm (Figure 1e,f; see [Supporting Information Figure S5b,c](#)). To the best of our knowledge, this system is the first kinetically controlled phthalocyanine aggregate exhibiting superlarge redshifted absorption.

The distinct morphological and spectral properties of the two nanostructures indicated that the molecular aggregation patterns of these two states were completely different. As further confirmation, the circular dichroism (CD) spectra of both PFI and PFII were measured

(see [Supporting Information Figure S6](#)). Both the Q-band and Soret band of PFII exhibited a positive to negative signal with increasing energy, suggesting a positive Cotton effect that referred to a right-handed helical structure.³⁵ This finding was consistent with the AFM characterization results. By contrast, PFI exhibited weak CD signals, indicating its low degree of order. Because the optical activity of the PF nanostructures was mainly derived from supramolecular chirality, while the intrinsic optical activity of the PF molecule was extremely low (not observed in a CD spectrum), the supramolecular structure of PFII was believed to be much more ordered than that of PFI.

In addition to the CD measurements, characterization by X-ray powder diffraction (XRD) and high-resolution transmission electron microscopy (HR-TEM) further revealed the different crystal structures of PFI and PFII (see [Supporting Information Figure S7](#)). The XRD results revealed that PFI had no crystallized structure but instead appeared rather amorphous. By contrast, the spectrum of PFII showed a significant peak of $d_{(002)} = 11.4$ Å at $2\theta = 7.78^\circ$, indicating the α -form of the phthalocyanine crystal structure.^{36–38} In addition, the PFII spectrum also showed a weak peak of $d = 3.4$ Å at $2\theta = 26.98^\circ$, indicating the stacking distance between neighboring phthalocyanine groups.³⁹ The HR-TEM image of PFII clarified that PFII had a well-ordered crystalline lattice parallel to the fibril growth direction. The corresponding fast Fourier transform pattern (see inset in [Supporting Information Figure S7b](#)) clearly showed individual spots that corresponded to the (002) lattice with 1.17 nm, which was consistent with the powder XRD result.

Because crystal structures generally have higher thermodynamic stability than disordered structures, it was reasonable to predict that PFII was thermodynamically more stable than PFI. In order to prove this assumption, a kinetic process was designed to compare the stability of these two nanostructures. Similar to the initial condition for preparing PFII, 0.25 mL of DMSO was added to 1.0 mL of PFI solution (0.1 mg mL^{-1}) to increase the ratio of DMSO/water solvent mixture. Over 4 days, PFI spontaneously transformed into PFII, accompanied by a decrease in the peak at 675 nm and the emergence of a near-infrared peak at 780 nm in the absorption spectra (Figure 2a,b). After removing DMSO via repeated dialysis, the absorption spectrum of the converted nanostructure remained unchanged. As the thermodynamic environment before and after the transformation process was completely the same (concentration, ionic strength, pH, temperature, etc.), it could be concluded that PFII was thermodynamically more stable than PFI. Furthermore, the time evolution of the transformation from PFI to PFII displayed sigmoidal profiles characterized by a short initial lag period corresponding to early stage nucleation followed by rapid elongation of the fibril

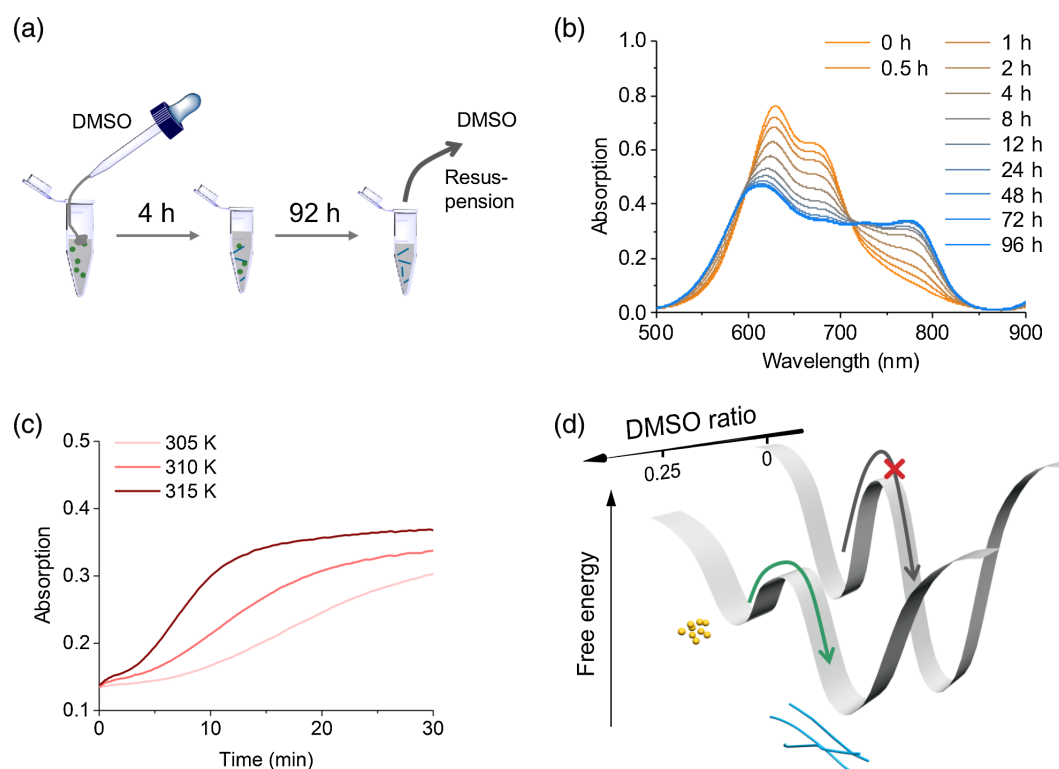


Figure 2 | (a) Schematic illustration of the conversion process from PFI to PFII, (b) spectral variation from PFI to PFII over time, (c) extinction kinetic profiles of the conversion process at different temperatures ($\lambda = 780$ nm), and (d) schematic representation of the free energy landscapes of PF assemblies at high (front) and low (back) DMSO/water volume ratios.

(Figure 2c), where different temperatures were set to accelerate the conversion.^{40,41} The conversion process images captured by TEM showed that the early stage of fibril nucleation was accompanied by nanoparticle collapse (see Supporting Information Figure S8).

These results obtained above further revealed that kinetic control over the assembly process played a crucial role in determining the morphology of the PF assemblies. For PFI, while nanoparticle formation is fast in a short period of time, the phthalocyanine-peptide conjugates tended to aggregate into spheres with the smallest surface to reduce interfacial tension. The nanoparticles had not reached their thermodynamic equilibrium, and the molecules in the assembly structures were randomly oriented. This is also an entropy-dominated process. In contrast, for PFII, due to the elongated assembly time and improved solubility, the PF molecules obtained sufficient freedom to adjust their intermolecular conformation and reach their thermodynamically more favored supramolecular structure: helical fibrils. This is an enthalpy-dominated process. The conversion from PFI to PFII proceeds when adding an extra amount of DMSO because DMSO is a good solvent for dissolving PF and reduces the free energy barrier between PFI and PFII (Figure 2d). The free energy barrier likely originates from the collapse of PFI and nucleation of PFII.

As a consequence, rapid nanoprecipitation as a general kinetic control without precisely tuning intermolecular structures leads to metastable nanoparticles, while a specific kinetic assembly method that enables tuning the intermolecular conformation leads to nanofibrils that are thermodynamically more stable.

To gain deeper insights into the assembly structures and elucidate the assembly processes, we performed computational simulations of both PFI and PFII via molecular dynamic simulation. Initially, all the inserted PF molecules were randomly distributed with arbitrary orientations in a periodic box. For PFI, the solvent was pure water. Over time, the PF molecules quickly assembled into a sphere-like supramolecular structure in 30 ns (Figure 3a). The assembly structure was disordered as an integer but showed a locally ordered substructure, the *H*-aggregated dimer, corresponding to the blueshifted absorption at 630 nm of PFI. By contrast to PFI, the simulation of PFII started in a mixed solvent of DMSO and water with a volume ratio of 1:2 (molar ratio 1:8). During the assembly process, an ordered supramolecular structure could be extracted, as shown in Figure 3b, c; the total assembly structure is shown in Supporting Information Figure S9. The phthalocyanine groups were stacked face-to-face with a slight stagger along one dimension to form a fibril with each two adjacent phthalocyanines,

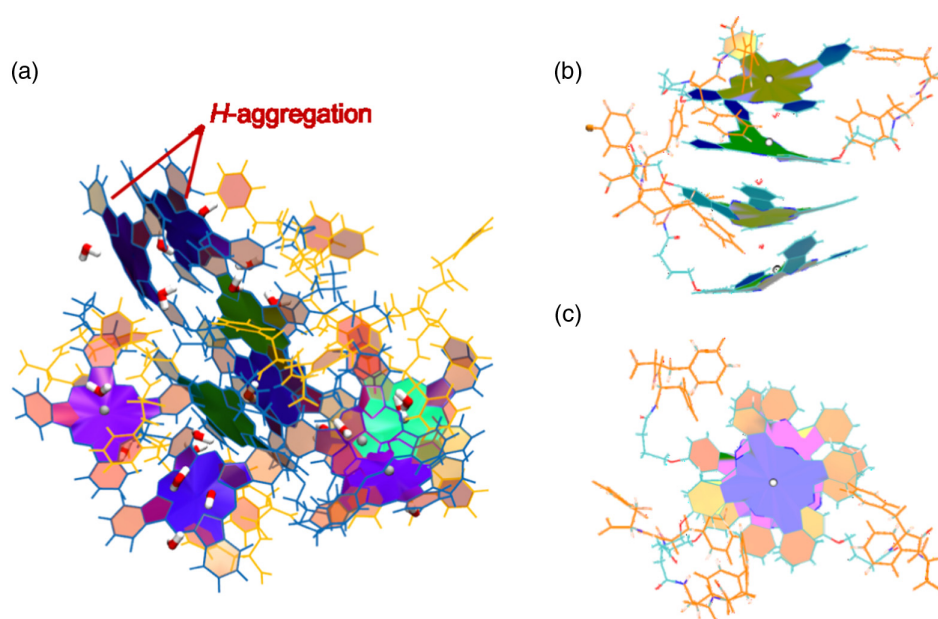


Figure 3 | Molecular dynamic simulations of PFI and PFII assembly structures. (a) The total PFI assembly structure with local H-aggregation pattern, (b) extracted ordered assembly structure of PFII, and (c) the same as (b), top view.

which were similar to the *J*-aggregation pattern. The interactions of such stacked phthalocyanine, strong π - π intermolecular interactions, are much stronger than those of phthalocyanines in a T-shape or irregular stacking state.⁴² PFII is thus shown to be more thermodynamically stable than PFI.

With the intermolecular conformations of PFI and PFII known, the origin of superlarge redshifted absorption could be determined. Conventionally, red- and blue-shifts in absorption originate from *J*- and *H*-aggregation, respectively, based on Kasha's perturbation theory.⁴³ This theory assumes that neighboring chromophores with a large intermolecular distance are coupled through only electrostatic interactions (or more precisely, dipole-dipole interactions), and that the excitation energy (E_{S1}) is perturbed only on a small scale (<20 nm shift of absorption). However, when neighboring chromophores are ordered and close enough that their molecular orbitals overlap, intermolecular charge transfer (CT) is allowed (Figure 4a).^{44,45} The CT state has much higher energy (E_{CT}) than the excitation energy, so it significantly affects the absorption. For PFII nanofibrils, the phthalocyanine groups were stacked in a slightly staggered pattern with a 3.4-Å intermolecular distance, which was estimated by density functional theory calculations and was consistent with the XRD result (Figure 4b). Such a short distance allowed effective CT and resulted in large splits in both the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) energies, ΔE_{H} , ΔE_{L} , and $\Delta E_{\text{L}+1}$ (where H is short for HOMO and L is short for LUMO) in comparison with those of the phthalocyanine monomer, as revealed by the

charge decomposition analysis diagram⁴⁶ (Figure 4c). The phthalocyanine dimer had ΔE_{L} and $\Delta E_{\text{L}+1}$ corresponding to two sets of coupled LUMOs because the phthalocyanine monomer had two near-degenerate LUMOs. Moreover, the orbital energy splitting is crucially dependent on the intermolecular distance of the neighboring phthalocyanines. Figure 4d reveals the dependence of orbital energy splitting on the intermolecular distance, where the energy splitting is represented by the CT integrals, t_{h} and t_{e} , following the relationships $-t_{\text{h}} \equiv \langle \phi_{\text{H}}^1 | \hat{h} | \phi_{\text{H}}^2 \rangle = 2\Delta E_{\text{H}}$ and $t_{\text{e}} \equiv \langle \phi_{\text{L}}^1 | \hat{h} | \phi_{\text{L}}^2 \rangle = 2\Delta E_{\text{L}}$, in which $\hat{h}$ is the single-electron Hamiltonian operator.⁴⁷ Accordingly, as CT is a short-range interaction, both t_{e} and t_{h} decreased dramatically with increasing distance. When the distance was larger than 5 Å, the effect of CT could be omitted. In contrast, for a very short distance lower than 3 Å, CT exhibited an extremely large effect on the orbital energy, which was advantageous for the superlarge redshift. However, the binding energy of the two chromophores increased rapidly, prohibiting very short distances. To reveal the result of CT effect, the dependence of the spectral variance on the chromophore distance was simulated (Figure 4e). Notably, for the short-distance region, the split of absorption in the Q-band region of phthalocyanine is significant. As there were two t_{e} with opposite signs, the electron transition of $\text{H} \rightarrow \text{L}$ with $t_{\text{h}}t_{\text{e}} < 0$ led to a large redshift, whereas $\text{H} \rightarrow \text{L}+1$ with $t_{\text{h}}t_{\text{e}} > 0$ led to a blueshift. As $\Delta E_{\text{L}} > \Delta E_{\text{L}+1}$, the redshift is larger than the blueshift, strongly consistent with the experimental results.

The kinetic control over the assembly not only affects the absorption spectrum but also tunes the emission of

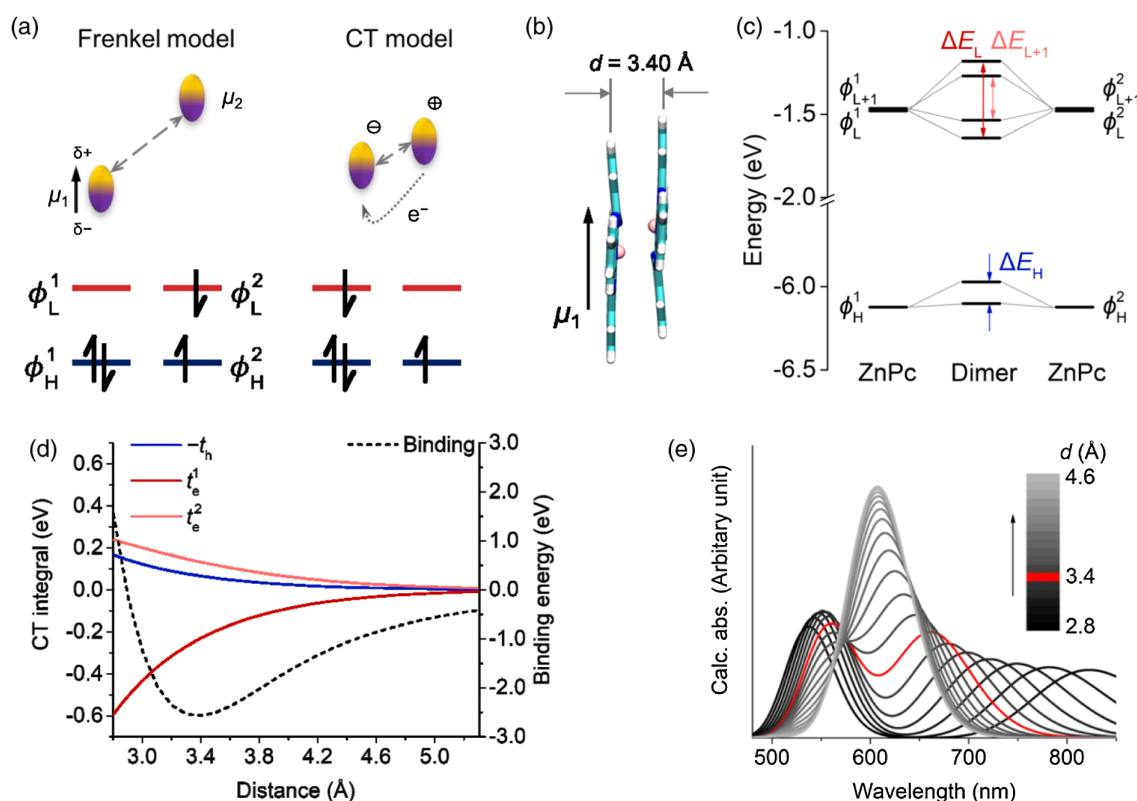


Figure 4 | Mechanism of superlarge redshifted absorption. (a) Comparison of the Frenkel model and CT model, where μ refers to the excited transition dipole moment; (b) the stacking zinc phthalocyanine dimer model of PFII, (c) orbital diagram and charge decomposition analysis of the dimer, (d) CT integrals as functions of intermolecular distance, and (e) calculated absorption spectra with different intermolecular distances.

the phthalocyanine-peptide conjugate nanostructures. The DMSO solution of PF showed significant fluorescence at 696 nm with a Stokes shift of 425 cm^{-1} (see Supporting Information Figure S4). In comparison, the fluorescence of PFI was completely quenched in both the visible and near-infrared regions even under laser irradiation. Different from PFI, PFII exhibited a fluorescence maxima at 820 nm with a Stokes shift of 630 cm^{-1} , indicating an assembly structure-dependent fluorescence (see Supporting Information Figure S10). The quenching of fluorescence in PFI was probably because the molecules were randomly assembled, and the fluorescence could be sufficiently quenched through vibrational relaxation. In contrast, in PFII, the molecules were regularly assembled, and the intramolecular vibration was inhibited to a certain degree. As vibrational relaxation and luminescence are competitive de-excitation processes, the fluorescence of PFII was not completely quenched.

In summary, starting from a model phthalocyanine-peptide conjugate building block, a novel *J*-aggregation state of phthalocyanine with a larger than 100-nm absorption redshift was successfully obtained by kinetic control over the self-assembly process for the first time. This kinetically controlled assembly resulted in nanofibrils with an ordered intermolecular arrangement that

was completely different from the nanoparticles created from rapid precipitation without tight regulation of the intermolecular interaction or conformation. Notably, the kinetically controlled nanofibrils were thermodynamically more stable than the nanoparticles, and sufficient intermolecular rearrangement was crucial for the formation of the nanofibrils. The superlarge redshifted absorption of the nanofibrils was further revealed to originate from the ordered stacking of the phthalocyanine molecules, in which considerable orbital overlap between neighboring chromophores generated CT states and thus significantly perturbed the excitation energy. The superlarge redshifted absorption induced by kinetically controlled assembly not only opens a new route toward easy fabrication of near-infrared materials but also helps reveal general design principles with greater promise for applications.

Supporting Information

Supporting information is available.

Conflicts of Interest

The authors declare no competing financial interest.

Acknowledgments

The authors acknowledge financial support from the National Natural Science Foundation of China (Project No. 21703252, 21522307, and 21473208), the National Natural Sciences Fund BRICS STI Framework Programme (51861145304) and Innovation Research Community Science Fund (No. 21821005), the Talent Fund of the Recruitment Program of Global Youth Experts, the Key Research Program of Frontier Sciences of the Chinese Academy of Sciences (CAS, Grant No. QYZDB-SSW-JSC034), and the CAS President's International Fellowship Initiative (2018VEA0005). The authors are also grateful to Dr. Dongdong Qi at University of Science and Technology Beijing for a grant of computer time.

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